# Question ID: fa2771d5

| Assessment | Test | Domain | Skill | Difficulty |
| --- | --- | --- | --- | --- |
| SAT | Math | Geometry and Trigonometry | Circles | Hard |

**Question**

Circle A has equation $(x-7)^2+(y+3)^2=1$. In the $xy$-plane, circle B is obtained by translating circle A to the right $4$ units. Which equation represents circle B?

**Answer**

A. $(x-7)^2+(y+7)^2=1$

B. $(x-3)^2+(y+3)^2=1$

C. $(x-11)^2+(y+3)^2=1$

D. $(x-7)^2+(y-1)^2=1$

**Correct Answer: C**

**Rationale**

Choice C is correct. The equation of a circle in the *xy*-plane can be written as $(x-h)^2+(y-k)^2=r^2$, where the center of the circle is $(h, k)$ and the radius of the circle is $r$ units. It's given that circle A has the equation $(x-7)^2+(y+3)^2=1$, which can be written as $(x-7)^2+\left(y-(-3)\right)^2=1^2$. It follows that $h=7$, $k=-3$, and $r=1$. Therefore, the center of circle A is $(7, -3)$ and its radius is $1$ unit. If circle A is translated $4$ units to the right, the *x*-coordinate of the center will increase by $4$, while the *y*-coordinate and the radius of the circle will remain unchanged. Translating the center of circle A to the right $4$ units yields $(7+4, -3)$, or $(11, -3)$. Therefore, the center of circle B is $(11, -3)$. Substituting $11$ for $h$, $-3$ for $k$, and $1$ for $r$ into the equation $(x-h)^2+(y-k)^2=r^2$ yields $(x-11)^2+\left(y-(-3)\right)^2=1$, or $(x-11)^2+(y+3)^2=1$. Therefore, the equation $(x-11)^2+(y+3)^2=1$ represents circle B.

Choice A is incorrect. This equation represents a circle obtained by shifting circle A down, rather than right, $4$ units.

Choice B is incorrect. This equation represents a circle obtained by shifting circle A left, rather than right, $4$ units.

Choice D is incorrect. This equation represents a circle obtained by shifting circle A up, rather than right, $4$ units.